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Course :- Operating Systems
Code :- CSA-0407

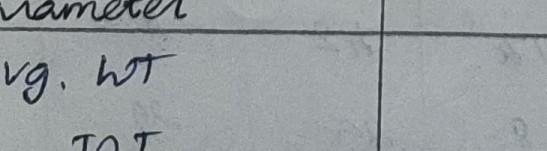
CO2 - AT1

1.) CPU Scheduling = Waiting Time & Turnaround time analysis.

Given,

Process	AT	BT	CT	TAT	WT
P_1	0	10	22	22	12
P_2	0	4	6	6	2
P_3	0	6	12	12	6
P_4	0	2	2	2	0

a.) Gantt chart,



TAT = CT - AT
WT = TAT - BT

b.) Avg. Waiting and Turnaround time,

$$\text{Avg. WT} = \frac{12+2+6+0}{4} = 5\text{ms}$$

$$\text{Avg. TAT} = \frac{22+6+12+2}{4} = 10.5\text{ms}$$

c.) Comparison of SJF and FCFS,

order :- $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

FCFS W.T, $P_1 = 0\text{ms}$, $P_2 = 10\text{ms}$, $P_3 = 14\text{ms}$, $P_4 = 20\text{ms}$

$$\text{Avg. WT} = \frac{0+10+14+20}{4} = 11\text{ms}$$

Parameter	FCFS	SJF
Avg. WT	11 ms	5 ms
Avg. TAT	16.5ms	10.5ms
CPU utilization	100 %	100 %

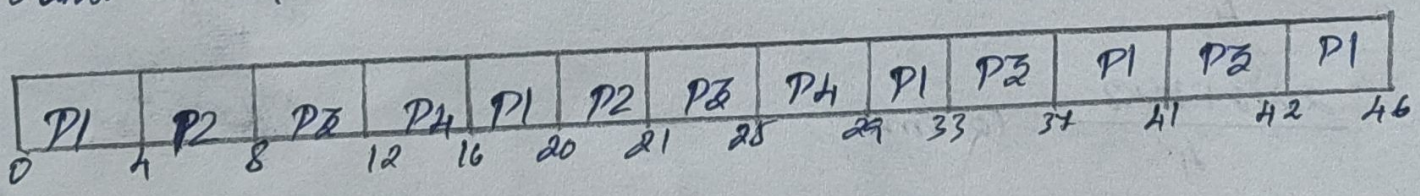
2.) Round robin scheduling - Time Quantum Impact:
Given,

Time quantum = 4ms

a.) Execution sequence

Execution	Process	Time Interval	Remaining BT
1	P1	0-4	16
2	P2	4-8	1
3	P3	8-12	9
4	P4	12-16	4
5	P1	16-20	12
6	P2	20-21	0
7	P3	21-25	5
8	P4	25-29	0
9	P1	29-33	8
10	P3	33-37	1
11	P1	37-41	4
12	P3	41-42	0
13	P1	42-46	0

Gantt chart:



b.) Calculate W.T and TAT,

Since AT = 0, TAT equals CT.

Completion times, P1 = 46ms, P2 = 21ms, P3 = 42ms, P4 = 29ms

W.T, P1, WT = 46 - 20 = 26

P2, WT = 21 - 5 = 16

P3, WT = 42 - 13 = 29

P4, WT = 29 - 8 = 21

Process	BT	CT	TAT	WT
P1	20	46	46	26
P2	5	21	21	16
P3	13	42	42	29
P4	8	29	29	21

Avg. WT = $\frac{92}{4}$ = 23ms

Avg. TAT = $\frac{138}{4}$ = 34.5ms

c.) Impact of Increasing Time Quantum to 10ms,

Order,

P1 → P2 → P3 → P4 → P1 → P3

New gantt chart,

P1	P2	P3	P4	P1	P3

Parameter	Quantum 4 ms	Quantum 10ms
Execution segments	17	6
Context switches	12	5
Avg. WT	23ms	22.25ms
Avg. TAT	34.5ms	33.75ms

2.) Deadlock – Safety check using Banker's algorithm.

Given,

(A, B, C) = (10, 5, 7)

a.) Need matrix,

Allocation,

Process	A	B	C
P1	0	1	0
P2	2	0	0
P3	3	0	2
P4	2	1	1
P5	0	0	2

Maximum,

Process	A	B	C
P1	7	5	3
P2	3	2	2
P3	9	0	2
P4	4	2	2
P5	5	3	3

Need = Maximum – Allocation

P1, (7, 5, 3) – (0, 1, 0) = (7, 4, 3)

P2, (3, 2, 2) – (2, 0, 0) = (1, 2, 2)

P3, (9, 0, 2) – (3, 0, 2) = (6, 0, 0)

P4, (4, 2, 2) – (2, 1, 1) = (2, 1, 1)

P5, (5, 3, 3) – (0, 0, 2) = (5, 3, 1)

Need matrix,

Process	A	B	C
P1	7	4	3
P2	1	2	2
P3	6	0	0
P4	2	1	1
P5	5	3	1

b) Determine whether the system is in safe state,

Calculate, A = 0+2+3+2+0 = 7

B = 1+0+0+1+0 = 2

C = 0+0+2+1+2 = 5

Total, (7,2,5)

Available = Total - Allocated

= (10,5,7) - (7,2,5) = (3,3,2)

Need ≤ Available

Process	Available before	Allocation released	Available after
P2	(3,3,2)	(2,0,0)	(5,3,2)
P4	(5,3,2)	(2,1,1)	(7,4,3)
P5	(7,4,3)	(0,0,2)	(7,4,5)
P1	(7,4,5)	(0,1,0)	(7,5,5)
P3	(7,5,5)	(3,0,2)	(10,5,7)

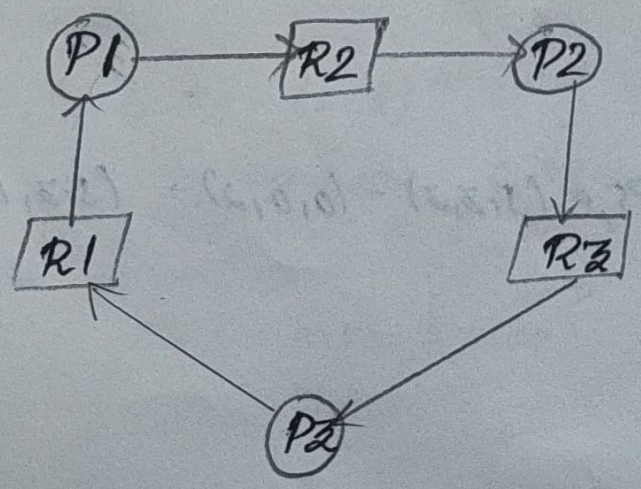
c) Valid safe sequence,

P2 → P4 → P5 → P1 → P3 //.

4) Deadlock detection - Resource allocation graph (RAG),

a) Resource allocation graph,

Graph representation;



Text representation;

P1 → R2 → P2

P2 → R3 → P3

P3 → R1 → P1

• Process (circle) ■ Resource (square)

b.) The graph contains a circular wait,

P1 → P2 → P3 → P1

result,

Deadlock exists.

c.) Recommended method,
Deadlock prevention using Resource Ordering.

fixed order, R1 < R2 < R3

advantages!.

1.) Prevents circular waiting.
2.) Reduces the possibility of deadlock.
3.) Provides a systematic resource allocation method.
4.) Improves system reliability.

For the existing deadlock, the OS can terminate or roll back one selected process, reclaim its resources, and allow the remaining processes to continue.

5.) CPU utilization – Multiprogramming Level Analysis!.
Given,
CPU execution time = 20%
I/O waiting time = 80%
Degree of multiprogramming = 4 processes.

CPU utilization,

$$U = 1 - p^n.$$

where, $p = 0.8$

a.) $p = 0.8$, $n = 4$

Substitute,

$$U = 1 - (0.8)^4$$
$$= 1 - 0.4096 = 0.5904$$

into percentage,

$$U = 0.5904 \times 100$$
$$= 59.04\%.$$

b.) Calculate CPU utilization for 6 processes.

$p = 0.8, n = 6$

Substitute,

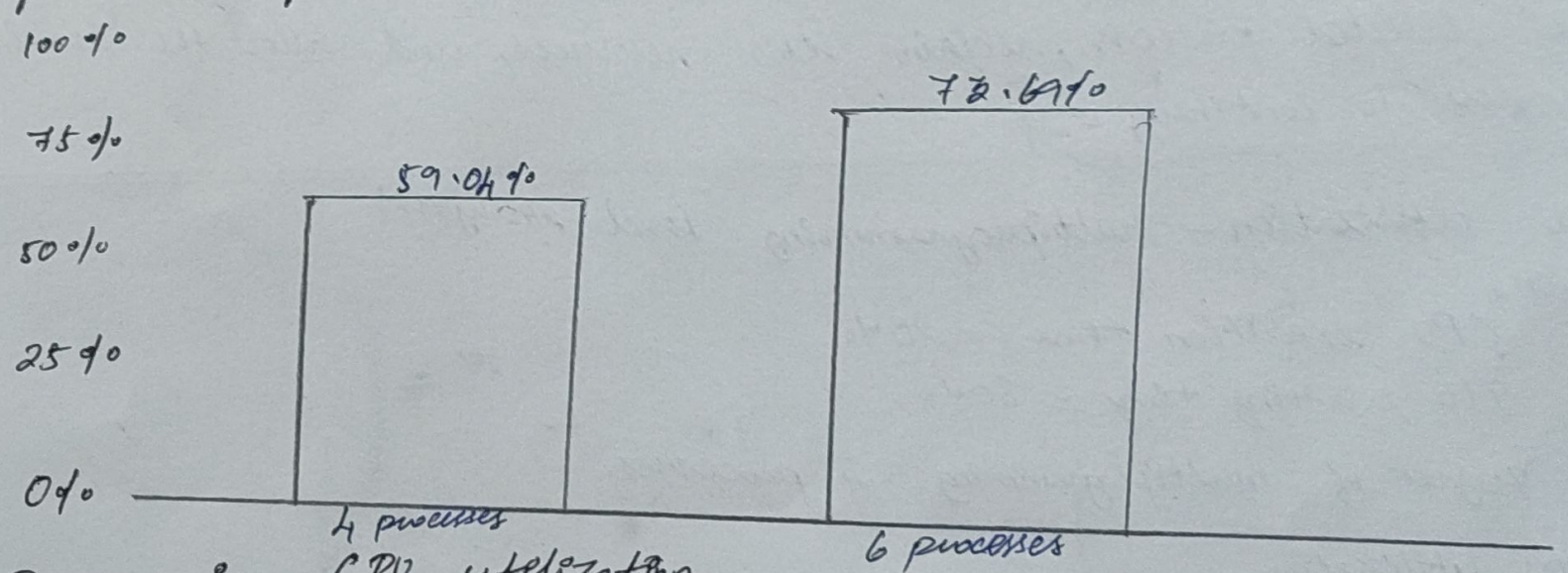
$$U = 1-(0.8)^6$$
$$= 1-0.262144 = 0.737856$$

$$\boxed{U = 73.79\%}$$

c) Compare and interpret system performance,

Parameter	4 processes	6 processes
I/O waiting probability	80%	80%
CPU utilization	59.04%	73.79%
CPU Idle probability	40.96%	26.21%
performance	Lower	Higher

graphical comparison,



Increase in CPU utilization,

$73.79 - 59.04 = 14.75$

It increases by approximately 14.75 percentage points.

Interpretation,

1.) Increasing multiprogramming allows more processes to remain in memory.

2.) When one process waits for I/O, another process can use the CPU.

3.) this reduce CPU idle time and improves the resource utilization.